



## 04 - DARS

# Models bilan ishlash, User model yaratish

Model · ORM · ForeignKey · migrate · admin

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## Dars haqida

Bu darsda ma'lumotlar bazasi bilan ishlashni boshlaymiz. Django **modellari** yordamida Python klassi orqali jadval yaratamiz — bu **ORM** deyiladi (SQL yozmaymiz).

### Dars yakunida siz quyidagilarni bilasiz:

- Model nima va ORM qanday ishlaydi;
- User va Blogs modellarini yozish;
- ForeignKey bilan bog'lash;
- makemigrations va migrate;
- admin panel va createsuperuser.

## 1. Model yozish

myapp/models.pyda ikki model yaratamiz. Har bir model — bazadagi bitta jadval, har bir maydon — ustun:

```
from django.db import models

class User(models.Model):
    name = models.CharField(max_length=100)
    surname = models.CharField(max_length=100)
    age = models.IntegerField()

    def __str__(self):
        return f'{self.name} {self.surname}'

class Blogs(models.Model):
    title = models.CharField(max_length=200)
    content = models.TextField()
    author = models.ForeignKey(User, on_delete=models.CASCADE)

    def __str__(self):
        return self.title
```

## 2. ForeignKey nima?

**ForeignKey** bir modelni boshqasiga bog'laydi. Bu yerda har bir **Blogs** qaysidir **Userga** tegishli (bittadan-ko'pga aloqa). `on_delete=CASCADE` — user o'chsa, uning bloglari ham o'chadi.

## 3. Migration: bazaga o'tkazish

```
python manage.py makemigrations
python manage.py migrate
```

### Ikkisining farqi:

- **makemigrations** — o'zgarishlardan 'reja' tuzadi;
- **migrate** — rejani bazaga qo'llaydi (jadval yaratadi).

## 4. Admin panelga ulash

myapp/admin.pyda modellarni ro'yxatdan o'tkazamiz:

```
from django.contrib import admin
from .models import User, Blogs

admin.site.register(User)
admin.site.register(Blogs)
```

Admin foydalanuvchi yaratib, panelga kiramiz:

```
python manage.py createsuperuser
python manage.py runserver
# Brauzer: http://127.0.0.1:8000/admin/
```

## 5. \_\_str\_\_ metodi

`__str__` obyekt admin panelda qanday nom bilan ko'rinishini belgilaydi. Usiz 'User object (1)' ko'rinadi, u bilan esa 'Ali Valiyev' kabi tushunarli chiqadi.

## Amaliy topshiriq

### Mustaqil bajaring:

1. User va Blogs modellarini yozing;
2. makemigrations va migrate ishga tushiring;
3. createsuperuser orqali admin yarating;
4. admin orqali 2 ta User va 3 ta Blog qo'shing.

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